Chapter 5

Two-Person bargaining

5.1 Nash Bargaining Solution

Let X be the set of possible outcomes. Let U be the set of feasible expected utilities from X, which is assumed to be a compact and convex subset of $\mathbb{R}^2$. Let $d \in U$ be the disagreement point, which is the outcome if the two parties fail to reach an agreement. We assume that $\exists u \in U$ such that $u \gg d$ (i.e., there is some outcome that is strictly better than the disagreement point for both parties). A bargaining solution is a function F that maps (U, d) to a point $F(U, d) \in U$.

Requirement: Scale Invariance

Consider the transformation $(U, d) \mapsto (\bar{U}, \bar{d})$ such that

$$\bar{U} = \left\{ (\bar{u}_1, \bar{u}_2) \middle| \bar{u}_i = \alpha_i u_i + \beta_i \text{ for some } \alpha_i > 0, \quad (u_1, u_2) \in U \right\},$$
$$\bar{d}_i = \alpha_i d_i + \beta_i.$$

Then,

$$F_i(\bar{U}, \bar{d}) = \alpha_i F_i(U, d) + \beta_i.$$

Requirement: Efficiency

$\nexists u \in U$ such that $u \gg F(U, d)$.

Requirement: Symmetry

If U is symmetric (i.e., $(u_1, u_2) \in U \iff (u_2, u_1) \in U$) and $d_1 = d_2$, then

$$F_1(U, d) = F_2(U, d).$$

Requirement: Independence of Irrelevant Alternatives (IIA)

Consider (U, d) and $(\bar{U}, d)$ such that $U \subseteq \bar{U}$. If $F(\bar{U}, d) \in U$, then,

$$F(\bar{U}, d) = F(U, d).$$

Theorem 5.1

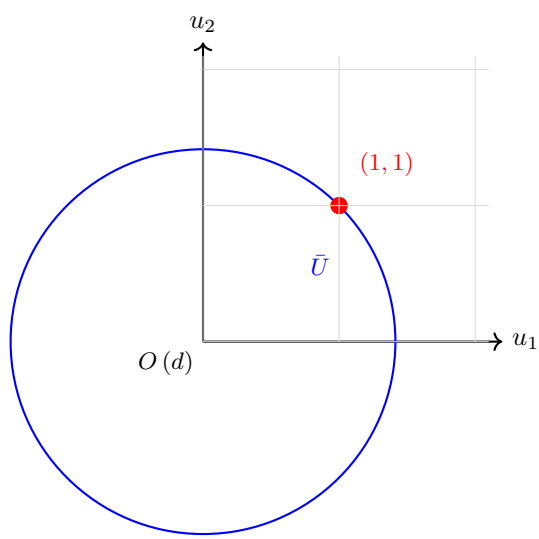
There is one and only one F that satisfies the above requirements. F is given by:

$$F^N(U, d) = \arg\max_{u \in U, u \geq d} (u_1 - d_1)(u_2 - d_2).$$

Proof for Theorem

Firstly, it is easy to check that F^N is well-defined in the sense that F^N satisfies the above requirements.

Let $(u_1^*, u_2^*) = F^N(U, d)$. We can always rescale the utilities such that $(u_1^*, u_2^*) \mapsto (1, 1)$ and $(d_1, d_2) \mapsto (0, 0)$, without loss of generality. Denote the new space as $(\bar{U}, \bar{d})$.



Claim

If we can show that $F = F^N$ for any F in the scaled space $(\bar{U}, \bar{d})$, then we must have $F = F^N$ for any (U, d).

Proof for Claim.

This is true by scale invariance.

Claim

$F^N(\bar{U}, d) = (1, 1)$.

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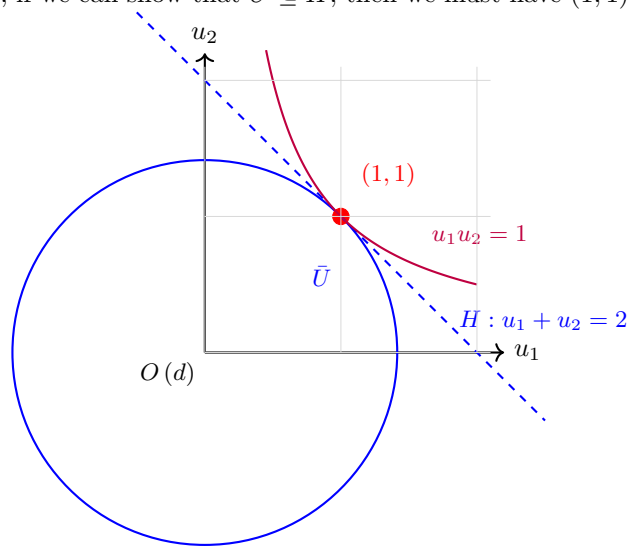
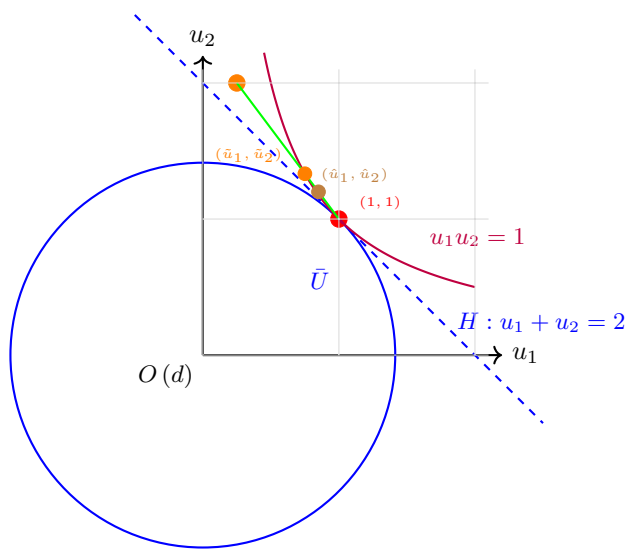
Proof for Claim.

Define the set $H = \{(u_1, u_2) \,|\, u_1 + u_2 \leq 2\}$. Note that the boundary of H is the tangent line to the hyperbola $u_1 u_2 = 1$ through $(1, 1)$ with slope -1. Since $(1, 1) \in \bar{U}$ and $(1, 1) \in H$, if we can show that $\bar{U} \subseteq H$, then we must have $(1, 1)$ as the solution.



To show that $\bar{U} \subseteq H$, suppose for contradiction that $\bar{U}$ contains some point (u_1, u_2) such that $u_1 + u_2 > 2$. The line segment connecting $(1, 1)$ and (u_1, u_2) must intersect the hypobola $u_1 u_2 = 1$ at some point $(\tilde{u}_1, \tilde{u}_2)$. So the line segment joining $(\tilde{u}_1, \tilde{u}_2)$ and $(1, 1)$ must be above the hypobola $u_1 u_2 = 1$. Since $\bar{U}$ is convex, we can find a convex combination of (u_1, u_2) and $(1, 1)$ that falls within the line segment joining $(\tilde{u}_1, \tilde{u}_2)$ and $(1, 1)$, denoted by $(\hat{u}_1, \hat{u}_2)$ such that $(\hat{u}_1, \hat{u}_2) \in \bar{U}$ and $\hat{u}_1 \hat{u}_2 > 1$. This contradicts the assumption that $F^N(\bar{U}, d) = (1, 1)$.



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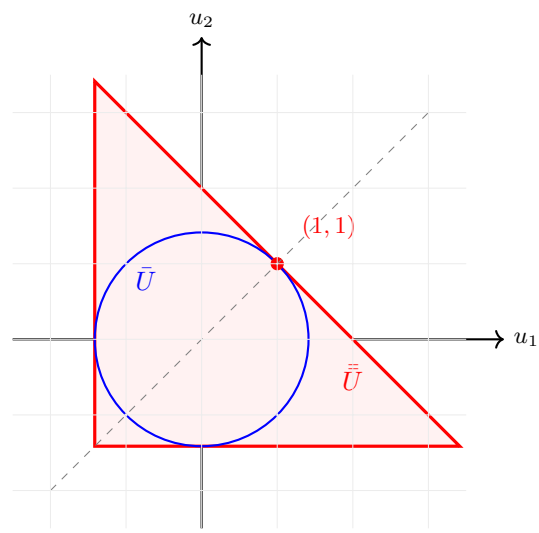
Claim

For any F, the solution must be $(1, 1)$.

Proof for Claim.

We can always include the $\bar{U}$ using a symmetric right triangle whose hypotenuse is on the boundary of H and whose right angle sits on the line of $u_1 = u_2$. Denote the new triangle by $\bar{\bar{U}}$.



By efficiency of F, the solution must lie on the boundary of $\bar{\bar{U}}$. By symmetry of F, the solution must be $(1, 1)$. Additionally, note that $\bar{U} \subseteq \bar{\bar{U}}$, and $F(\bar{\bar{U}}, \bar{d}) \in \bar{U}$. By IIA, we must have $F(\bar{\bar{U}}, \bar{d}) = F(\bar{U}, \bar{d})$. Therefore, $F(\bar{U}, \bar{d}) = (1, 1)$.

From all the results above, we conclude that $F = F^N$ for any (U, d).

Remark.

This is actually a *normative* result in the sense that we require the bargaining solution to satisfy certain axioms. The Nash bargaining solution is the unique solution that satisfies these axioms. However, it is not necessarily a positive result in the sense that we do not have a game-theoretic model that can generate the Nash bargaining solution as an equilibrium outcome.

5.2 Dividing a Dollar

Let $X = [0, 1]$. Suppose utility functions for the two players are $u_1 : X \to \mathbb{R}$ and $u_2 : X \to \mathbb{R}$, which are continuous, strictly increasing, and strictly concave. The problem is given by

$$\max_{x \in [0,1]} u_1(x) u_2(1 - x).$$

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The FOC is:

$$\begin{aligned} & u_1'(x^*)u_2(1-x^*) - u_1(x^*)u_2'(1-x^*) = 0 \\ \implies & \frac{u_1'(x^*)}{u_1(x^*)} = \frac{u_2'(1-x^*)}{u_2(1-x^*)}. \end{aligned}$$

Suppose in addition that Player 1 becomes more risk averse than Player 2:

$$\begin{aligned} & v_1 : X \to \mathbb{R}, \text{ where } v_1 = \phi(u_1), \quad \phi' > 0, \quad \phi'' < 0, \\ & u_2 : X \to \mathbb{R}. \end{aligned}$$

So the problem becomes:

$$\max_{x\in[0,1]} v_1(x)u_2(1-x) = \max_{x\in[0,1]} \phi(u_1(x))u_2(1-x).$$

The FOC is:

$$\begin{aligned} & \phi'(u_1(x^{**}))u_1'(x^{**})u_2(1-x^{**}) - \phi(u_1(x^{**}))u_2'(1-x^{**}) = 0 \\ \implies & \frac{\phi'(u_1(x^{**}))u_1'(x^{**})}{\phi(u_1(x^{**}))} = \frac{u_2'(1-x^{**})}{u_2(1-x^{**})}. \end{aligned}$$

Since ϕ is a concave function,

$$\phi'(t) < \frac{\phi(t)}{t} \implies \frac{\phi'(t)}{\phi(t)} < \frac{1}{t}.$$

Thus,

$$\frac{u_1'(x^*)}{u_1(x^*)} > \frac{\phi'(u_1(x^{**}))u_1'(x^{**})}{\phi(u_1(x^{**}))} = \frac{u_2'(1-x^{**})}{u_2(1-x^{**})}.$$

So in comparison,

$$\begin{cases} \frac{u_1'(x^*)}{u_1(x^*)} = \frac{u_2'(1-x^*)}{u_2(1-x^*)} \\ \frac{u_1'(x^{**})}{u_1(x^{**})} > \frac{u_2'(1-x^{**})}{u_2(1-x^{**})} \end{cases} \implies x^{**} < x^*.$$

5.3 Alternating Offer Model

Consider the problem of dividing a dollar: Two players will propose a division of a dollar in turns. Without loss of generality, suppose Player 1 proposes first. If Player 1 proposes to give x to Player 2, then Player 2 can either accept or reject. If Player 2 accepts, then the payoffs are $(1-x, x)$. If Player 2 rejects, then the game moves to the next period, where Player 2 proposes a division of a dollar and Player 1 can accept or reject. If Player 1 accepts, then the payoffs are $(x', 1-x')$. If Player 1 rejects, then the game continues to the next period, and so on. The game ends when one proposal is accepted. The discount factor for both players is $\delta \in (0,1)$.

Claim

Any split $(x, 1-x)$ where $x \in (0,1)$ is a NE outcome.

Proof for Claim.

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This split can be supported by the strategies where Player 1 only accepts proposals that give them at least x, and Player 2 only accepts proposals that give them at least $1 - x$.

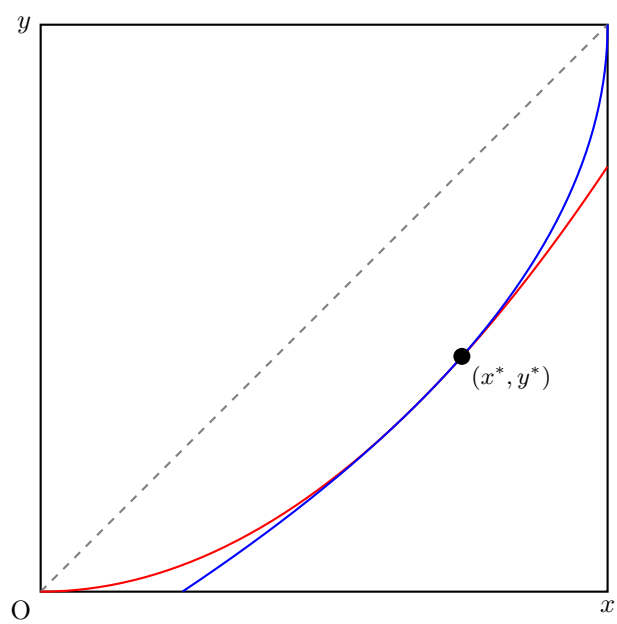
Theorem 5.2: Rubinstein

Let (x^*, y^*) be the solution to

$$u_1(y) = \delta u_1(x)$$
$$u_2(1-x) = \delta u_2(1-y)$$

Then there is a unique subgame perfect equilibrium in which in G_1, Player 1 asks for x^* and accepts iff $y \geq x^*$, and in G_2, Player 2 offers y^* and Player 1 accepts iff $x \leq y^*$.

Proof for Theorem



Let (x^*, y^*) be the solution to

$$u_1(y) = \delta u_1(x)$$
$$u_2(1-x) = \delta u_2(1-y)$$

Player 1 always offers x^*, and rejects Player 2's offer if $y < x^*$. Player 2 always offers y^*, and rejects Player 1's offer if $x > y^*$.

(z, t) is an outcome if a split of $(z, 1-z)$ is accepted in period t.

Define $v_1(z, t)$ as the dollar amount such that Player 1 is indifferent between receiving $v_1(z, t)$ today and receiving z in period t.

$$u_1(v_1(z, t)) = \delta^{t-1} u_1(z).$$

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Define $v_2(z,t)$ as the dollar amount such that Player 2 is indifferent between giving Player 1 $v_2(z,t)$ today and giving Player 1 z in period t.

$$u_2(1 - v_2(z,t)) = \delta^{t-1} u_2(1-z).$$

Define

$$M_i = \sup\{v_i(z,t) : (z,t) \text{ is a SPE of } G_i\},$$
$$m_i = \inf\{v_i(z,t) : (z,t) \text{ is a SPE of } G_i\},$$

where G_i is the subgame starting with Player i's proposal.

Claim

$$u_1(M_2) \leq \delta u_1(M_1).$$

Proof for Claim.

If Player 2 offers y such that $u_1(y) > \delta u_1(M_1)$, then Player 1 would accept the offer, because the right hand side $\delta u_1(M_1)$ is the most that Player 1 can get in the subsequent G_1. So Player 2 would never offer such y. Hence, we must have $u_1(M_2) \leq \delta u_1(M_1)$.

Claim

$$u_1(M_2) \geq \delta u_1(M_1).$$

Proof for Claim.

If Player 2 offers y such that $u_1(y) < \delta u_1(M_1)$, then Player 1 will reject the offer, because by rejecting and entering G_1, Player 1 can get at most M_1 (with current payoff equivalent $\delta u_1(M_1)$). Therefore, any successful offer in G_2 must satisfy $u_1(y) \geq \delta u_1(M_1)$. Taking the supremum over all y in G_2 equilibria, we have $u_1(M_2) \geq \delta u_1(M_1)$.

Claim

$$u_2(1 - M_1) \geq \delta u_2(1 - M_2).$$

Proof for Claim.

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In G_1, if Player 1 demands more than $v_2(M_2, 2)$, then Player 2 will reject. This implies

$$M_1 \leq v_2(M_2, 2).$$

And this implies

$$u_2(1 - M_1) \geq u_2(1 - v_2(M_2, 2)) = \delta u_2(1 - M_2).$$

5.4 Unequal Discounting

Suppose now Player 1 has utility function w_1 and discount factor δ_1, and Player 2 has utility function w_2 and discount factor δ_2, with $\delta_1 < \delta_2$. We make the following transformation of the original problem:

- Player 2 has $u_2 = w_2$ and $\delta = \delta_2$.
- Player 1 has $u_1 = [w_1(x)]^{\frac{\ln \delta_2}{\ln \delta_1}}$ and $\delta = \delta_2$. (Note that $\frac{\ln \delta_2}{\ln \delta_1} < 1$ so that u_1 is concave, and is even "more concave" than w_1.)

Without loss of generality, assume that

$$\begin{aligned}
& \delta_1^{t-1} w_1(z) > \delta_1^{t'-1} w_1(z') \\
\iff & \left[\delta_1^{t-1} w_1(z)\right]^{\frac{\ln \delta_2}{\ln \delta_1}} > \left[\delta_1^{t'-1} w_1(z')\right]^{\frac{\ln \delta_2}{\ln \delta_1}} \\
\iff & \frac{\ln \delta_2}{\ln \delta_1} \left[(t-1) \ln \delta_1 + \ln w_1(z)\right] > \frac{\ln \delta_2}{\ln \delta_1} \left[(t'-1) \ln \delta_1 + \ln w_1(z')\right] \\
\iff & (t-1) \ln \delta_2 + \frac{\ln \delta_2}{\ln \delta_1} \ln w_1(z) > (t'-1) \ln \delta_2 + \frac{\ln \delta_2}{\ln \delta_1} \ln w_1(z') \\
\iff & \ln \delta_2^{t-1} + \ln \left[w_1(z)\right]^{\frac{\ln \delta_2}{\ln \delta_1}} > \ln \delta_2^{t'-1} + \ln \left[w_1(z')\right]^{\frac{\ln \delta_2}{\ln \delta_1}} \\
\iff & \delta_2^{t-1} u_1(z) > \delta_2^{t'-1} u_1(z').
\end{aligned}$$

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